



Fifth Semester B.Tech. (Minor in IOT) (CBCS) Degree Examinations

January 2025

## EC3106-1 – CONNECTED DEVICES SECURITY

Duration: 3 Hours

Max. Marks: 100

**Note:**

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B: Descriptive Answer Questions:** Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

### PART - A: MULTIPLE CHOICE QUESTIONS

**20 Marks**

1. What is the main security challenge associated with IoT devices?  
A) Lack of device diversity  
B) High cost of deployment  
C) Limited processing power for security measures  
D) Easy integration with existing systems
2. Which of the following authentication methods is considered the most secure?  
A) Single layer password authentication  
B) Two factor authentication  
C) Biometric authentication  
D) No authentication
3. Which of the following is NOT a characteristic of IoT systems?  
A) Interconnectivity  
B) Scalability  
C) Robustness  
D) High power consumption
4. Which of the following best defines the Internet of Things (IoT)?  
A) A network of interconnected computers  
B) A global network connecting smart devices and sensors  
C) A collection of web applications  
D) A cloud computing service
5. Which of the following is a symmetric encryption algorithm?  
A) RSA  
B) DES  
C) ECC  
D) DSA
6. HMAC (Hashbased Message Authentication Code) combines which two elements?  
A) Hash function and encryption key  
B) Hash function and message  
C) Encryption key and message  
D) Public key and private key
7. Which mode of operation in block ciphers is considered less secure due to its lack of randomization?  
A) CBC (Cipher Block Chaining)  
B) ECB (Electronic Codebook)  
C) CFB (Cipher Feedback)  
D) OFB (Output Feedback)
8. The primary purpose of a digital signature is to  
A) Encrypt the message  
B) Verify the authenticity and integrity of the message  
C) Generate random numbers  
D) Compress data
9. Which of the following best describes a threat in the context of IoT security?  
A) A potential exploitation of a vulnerability  
B) A weakness in a system or network  
C) A measure taken to protect against attacks  
D) A method used to encrypt data
10. Which of the following is a method used to mitigate risks in IoT environments?  
A) Disabling encryption  
B) Using default passwords  
C) Implementing strong access control measures  
D) Ignoring firmware updates
11. Which of the following is considered a noninvasive attack on an IoT device?  
A) Power analysis attack  
B) Microprobing  
C) Fault injection  
D) Sidechannel attack
12. What type of attack involves directly tampering with the hardware of an IoT device?  
A) Logical attack  
B) Invasive attack  
C) Phishing attack  
D) Man in the middle attack

13. What is the main purpose of implementing threat mitigation strategies in IoT?  
 A) To reduce the impact and likelihood of threats B) To eliminate all possible threats  
 C) To increase system complexity D) To identify new features for IoT devices
14. Which of the following is NOT a common threat mitigation measure in IoT security?  
 A) Encryption of data B) Regular firmware updates  
 C) Using strong authentication mechanisms D) Disabling firewalls
15. Which of the following is a key component of a secure software development life cycle (SDLC)?  
 A) Code obfuscation B) Ignoring vulnerability reports  
 C) Regular security audits and code reviews D) Limiting user access
16. In threat modeling, which technique involves identifying possible threats based on the functionality of the system?  
 A) Asset analysis B) Threat vector mapping  
 C) Attack surface analysis D) Risk modeling
17. What is the primary function of a Trusted Execution Environment (TEE)?  
 A) To run general applications B) To provide a secure area for sensitive operations  
 C) To improve graphics processing D) To increase the speed of the device
18. Which technology is commonly used to create a secure environment within an ARM processor?  
 A) TrustZone B) Secure Boot  
 C) Full Disk Encryption D) Firewalls
19. What does TEE stand for in the context of platform security?  
 A) Trusted Execution Environment B) Technical Execution Engine  
 C) Total Encryption Environment D) Trusted Encryption Endpoint
20. Which security feature is commonly found in secured MCUs?  
 A) Inbuilt web browser B) Hardware based cryptographic engine  
 C) Wireless charging capability D) Larger storage capacity

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

| Unit – I  |                                                                                                                                                                               | Marks | BT* | CO* | PO* |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1.        | a) What are the components of Symmetric encryption? Explain monoalphabetic cipher by considering the key "ANOTHER" as an example.                                             | 8     | L*2 | 2   | 1   |
|           | b) Identify and explain the fundamental building blocks of IoT, emphasizing their roles in enabling IoT applications.                                                         | 8     | L2  | 1   | 1   |
| 2.        | a) Explain the significance of availability in IoT. What are the key considerations for ensuring availability in IoT? Discuss the challenges in ensuring availability in IoT. | 8     | L2  | 1   | 1   |
|           | b) What is symmetric cryptography and how does it relate to AES CBC and ECB mode?                                                                                             | 8     | L2  | 2   | 1   |
| 3.        | a) Define Cryptosystem. Explain Caesar cipher. Encrypt the message "do not disclose this message" using Caesar cipher encryption.                                             | 8     | L2  | 2   | 1   |
|           | b) Define Authentication. Discuss the importance and Challenges in IoT Authentication in brief.                                                                               | 8     | L2  | 1   | 1   |
| Unit – II |                                                                                                                                                                               |       |     |     |     |
| 4.        | a) What are True Random Number Generators (TRNG)? Discuss the components, key characteristics and security considerations for TRNG.                                           | 8     | L2  | 4   | 1   |
|           | b) Explain randomization software design patterns with detailed example that can be used to prevent side-channel attacks on IoT devices?                                      | 8     | L2  | 3   | 1   |
| 5.        | a) What are semi-invasive attacks in IoT security? Explain different methods of semi-invasive attacks. What are the techniques in preventing semi-Invasive Vulnerabilities.   | 8     | L2  | 4   | 1   |
|           | b) What is a threat model and how can it be used to identify potential threats to IoT devices?                                                                                | 8     | L2  | 3   | 1   |

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|-------|-------------------------------------------------------------------------------------------------|---|----|---|---|
| 6. a) | What are Physical Unclonable Function (PUF)? Discuss the key concepts and applications for PUF. | 8 | L2 | 3 | 1 |
| b)    | What are some device-level security concepts that can be used to secure IoT devices?            | 8 | L2 | 4 | 1 |

**Unit – III**

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|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| 7. a) | What are the security requirements of TEE and why is it considered to be more trustworthy than other security solutions? Provide a detailed explanation. | 8 | L2 | 5 | 4 |
| b)    | What are secured elements and how can they be used to enhance IoT security? Provide a detailed explanation.                                              | 8 | L2 | 5 | 1 |
| 8. a) | With the help of a neat block diagram, give the detailed explanation of TEE architecture.                                                                | 8 | L2 | 5 | 1 |
| b)    | What are secured MCUs? Explain the key security capabilities of a secured MCU.                                                                           | 8 | L2 | 5 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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